

Predicting Wildfire Ignition Cause in Canada

Using Machine Learning and Climate Data

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01 Motivation

Why does it matter?

Wildfires are one of Canada's most significant natural hazards. The **2023 season burned 16M+ hectares** — more than double any previous record.

Natural (lightning) fires are largely unavoidable. Human-caused fires are **entirely preventable** through targeted interventions.

Research gap: Modern ML for cause prediction with integrated climate data remains underexplored in the Canadian context.

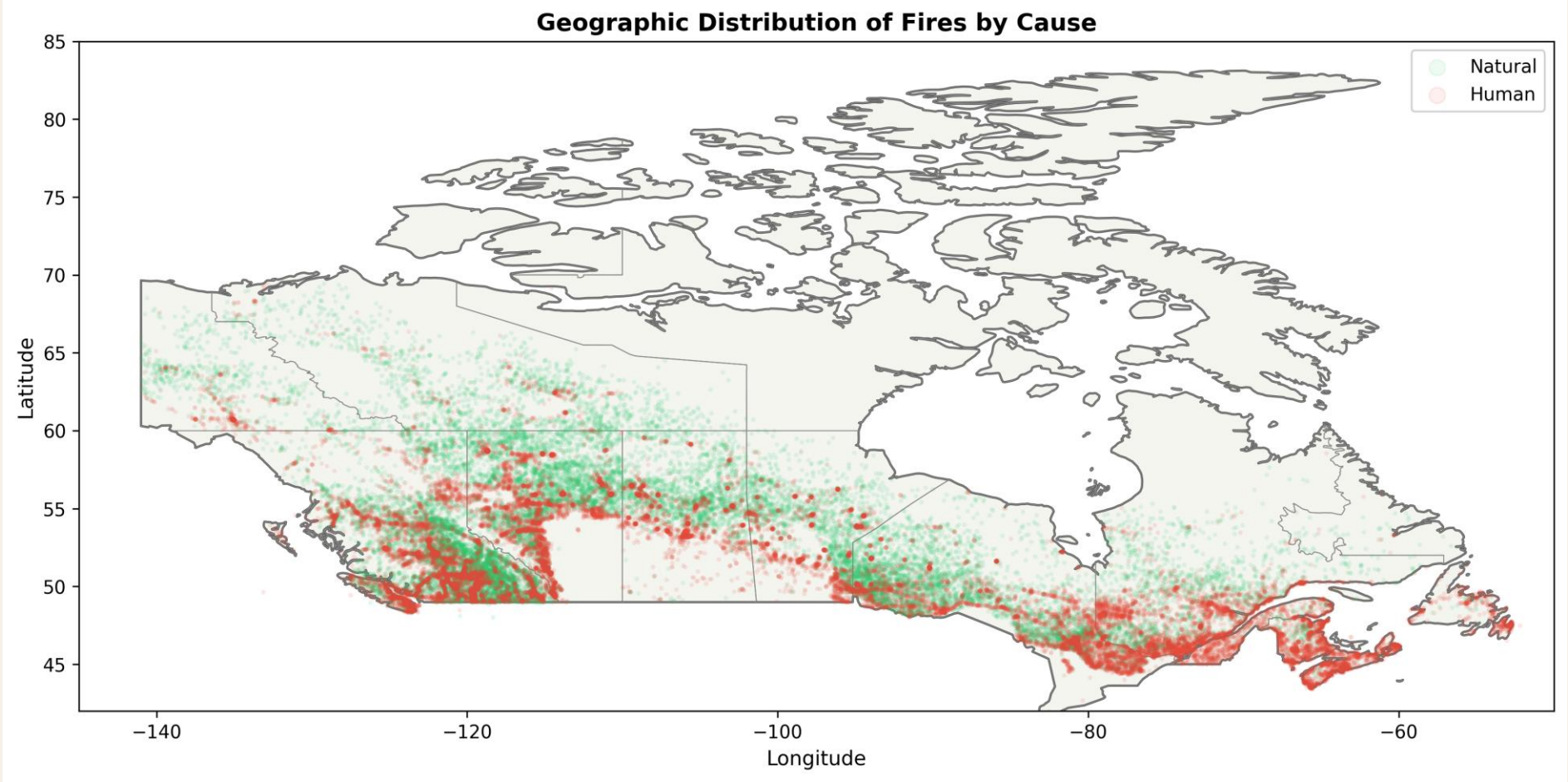


Fig. 1: Geographic distribution of fires by cause (50K sample)

374,037 fires · 45.9% Natural · 54.1% Human

02 Data Sources

Canadian National Fire Database (CNFDB)

442,403 raw records, 1972–2024 · Filtered to 374,037 fires with known cause labels and valid coordinates

Environment Canada Climate Data

231 weather stations · 12,243 daily CSV files
Max/min/mean temp, precipitation, wind, snow

Spatial Matching

Nearest-station haversine · Median distance **102 km**
Temp coverage 82.9% · Precip coverage 82.5%

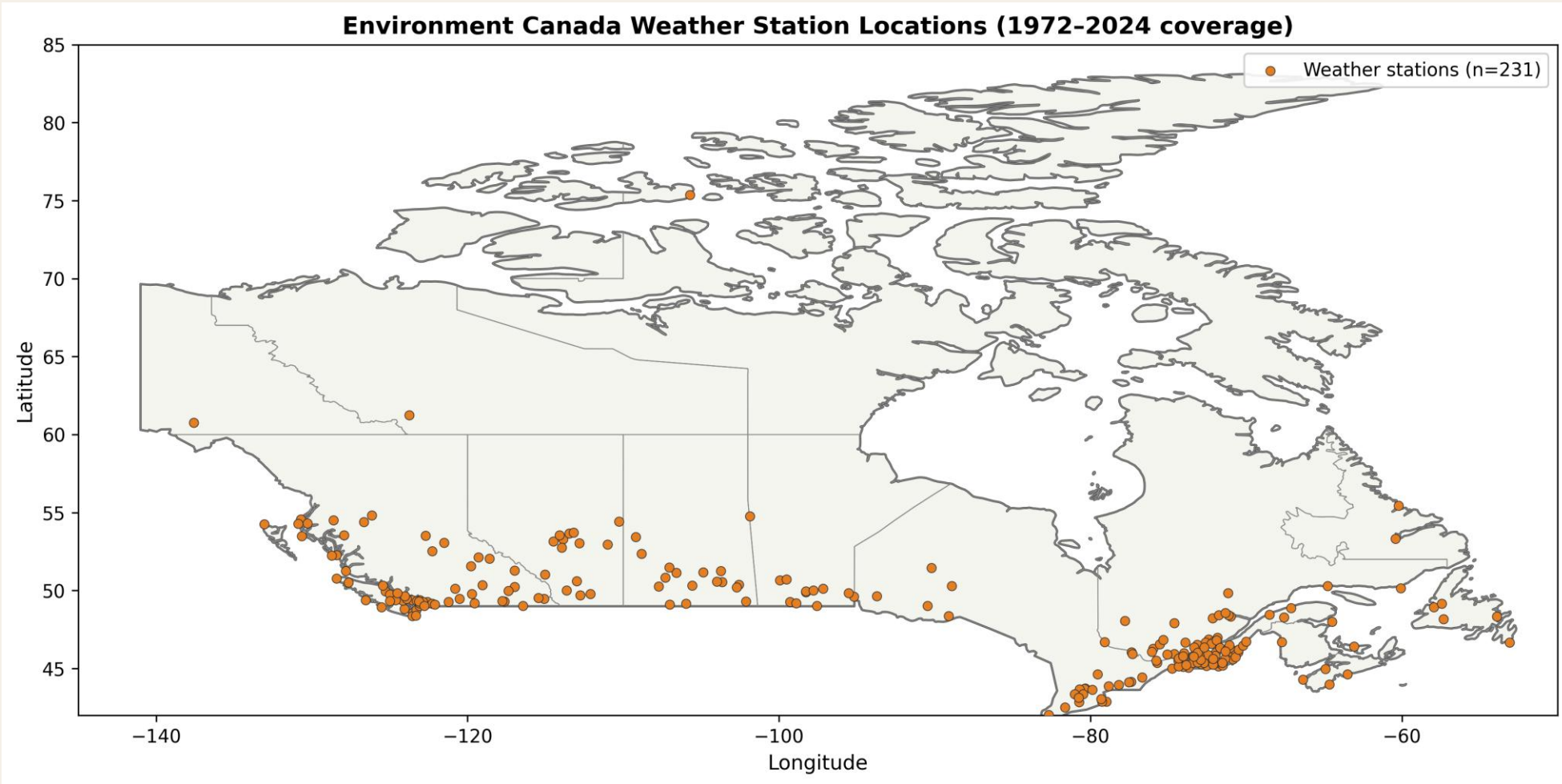


Fig. 2: 231 Environment Canada weather station locations (1972–2024)

03 Methodology

Feature Engineering (33 features)

Temporal (6): Day of year (sin/cos encoding for circularity), month, year, weekend flag
Geographic (3): Province (label-encoded), latitude, longitude

Fire (1): Log-transformed fire size — captures tendency for natural fires to grow larger before detection

Climate day-of (8): Max/min/mean temp, precipitation, rainfall, snow on ground, wind gust, diurnal temp range

Antecedent (15): 7-day and 30-day windows — mean/max temp, total precip, dry day count, temp range; plus temp anomaly and composite fire weather index

Model: Gradient Boosted Decision Tree

GBDT was chosen for its ability to handle mixed feature types, model non-linear interactions, natively tolerate missing values, and provide interpretable feature importance. Missing climate values encoded as –999 rather than imputed — GBDT learns optimal split directions for absent data.

500 estimators · Early stopping patience 20 · Learning rate 0.05
Max depth 5 · 80% row subsampling · sqrt(p) column subsampling
Inverse class-frequency sample weights

Evaluation

Stratified 80/20 train-test split (299,160 train / 74,790 test)

5-fold stratified cross-validation for stability

Permutation test (1,000 iterations) for significance

Bootstrap 95% confidence intervals (1,000 resamples)

Primary metric: PR-AUC — more informative than ROC-AUC under mild class imbalance

Key Performance Metrics — Held-Out Test Set (74,790 fires)

0.946

PR-AUC

Primary metric

0.932

ROC-AUC

Discrimination

0.859

F1-Score

Human class

0.853

Accuracy

Overall

0.709

MCC

Matthews Corr. Coef.

p<0.001

Significance

Permutation test

04 Classification Results

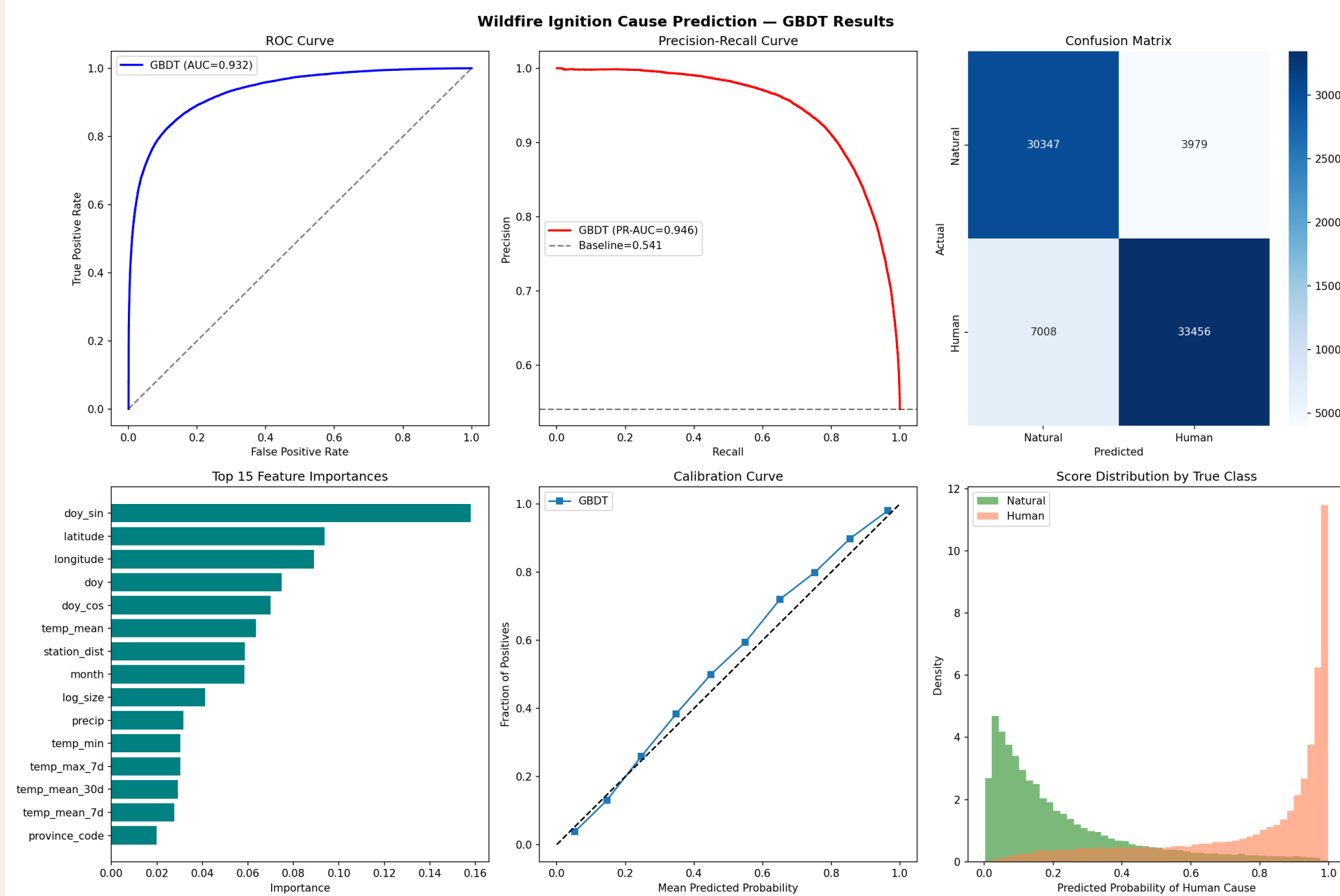


Fig. 3: ROC curve, PR curve, confusion matrix, calibration, score distributions

Class	Precision	Recall	F1	n
Natural	0.812	0.884	0.847	34,326
Human	0.894	0.827	0.859	40,464
Wtd Avg	0.856	0.853	0.853	74,790

5-fold CV: ROC-AUC 0.9324 ± 0.0003 (range 0.932–0.933)

Permutation test: $p < 0.001$ · Bootstrap 95% CI: [0.931, 0.934]

Brier score 0.104 · Log loss 0.333

Well-calibrated probabilities → suitable for real-time risk scoring.

05 Feature Importance

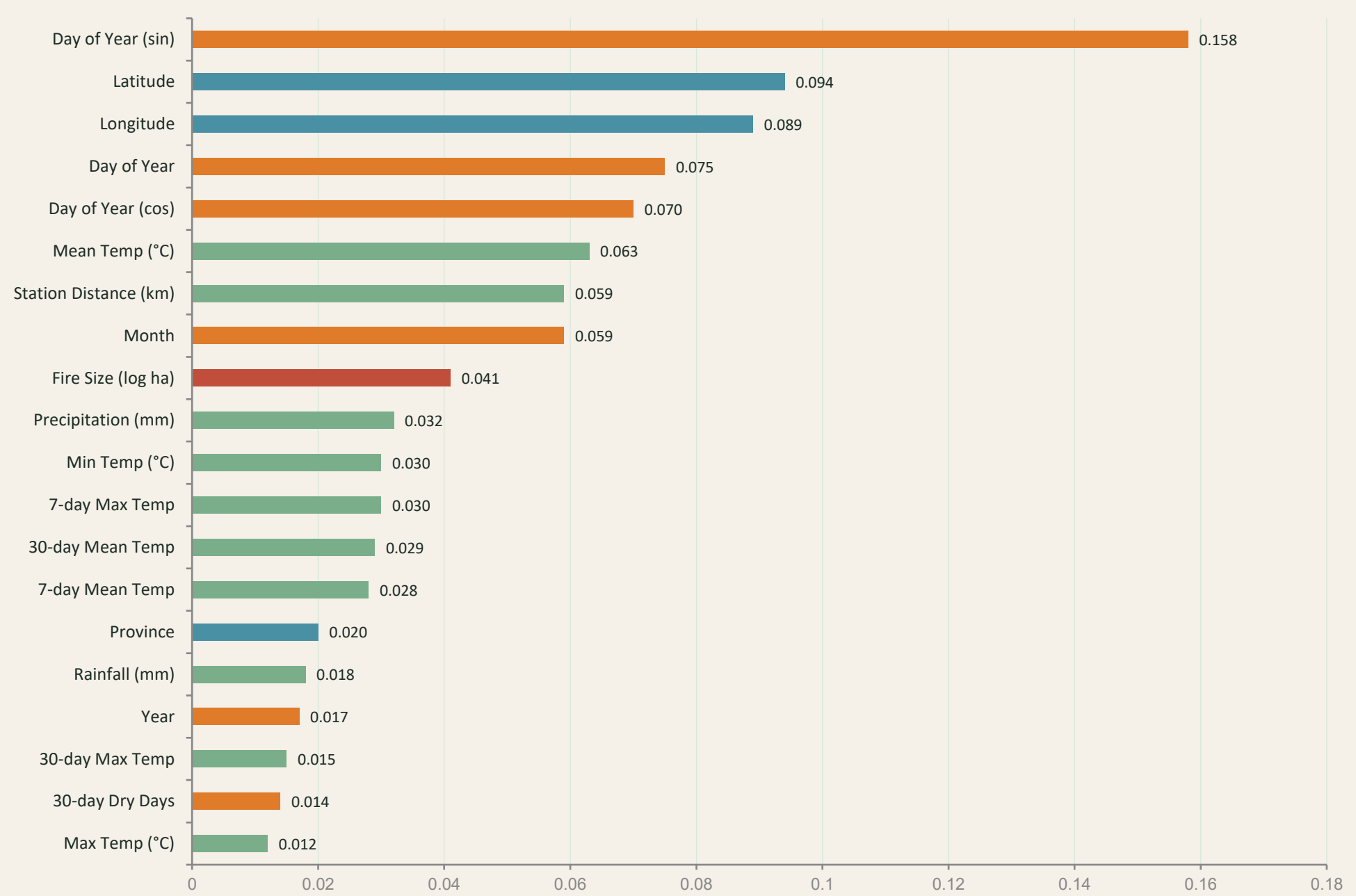


Fig. 4: Top 20 features by gain-based (Gini) importance | Color: orange = temporal, teal = geographic, red = fire size, green = climate

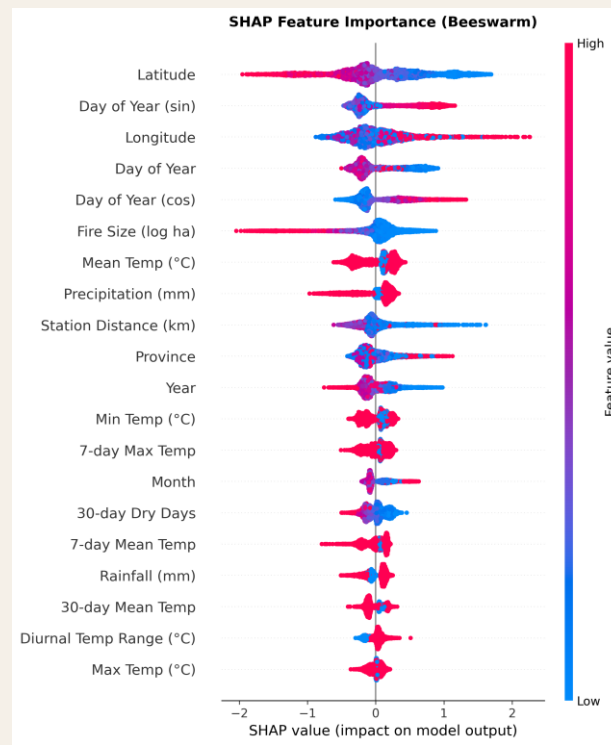


Fig. 5: SHAP beeswarm — direction & magnitude of feature effects on P(human cause)

06 Discussion & Conclusions

Ecological Interpretation

Temporal dominance aligns with fire science: **lightning storms peak July–August** in Canada's boreal forests. Human fires span a broader season — spring burns, summer camping, autumn land clearing.

Geographic signal reflects spatial segregation:

populated southern regions → human fires
remote boreal north → natural fires.

Smaller fires are more likely human-caused; larger fires more likely natural.

Implications for Fire Management

- Concentrate campaigns April–June (human fires peak)
- Redirect to detection & suppression in July–August
- Eastern provinces: awareness campaigns & fire bans
- Northern territories: early detection & remote response
- Calibrated probabilities enable dynamic risk scoring

Limitations

Climate features regional (median 102 km). No land use, road proximity, or vegetation type. Fire size requires post-ignition data. FWI not directly computed.

Future Work

FWI integration · Road & landscape features · Satellite NDVI · Spatial-temporal deep learning · Operational deployment with CWFI5

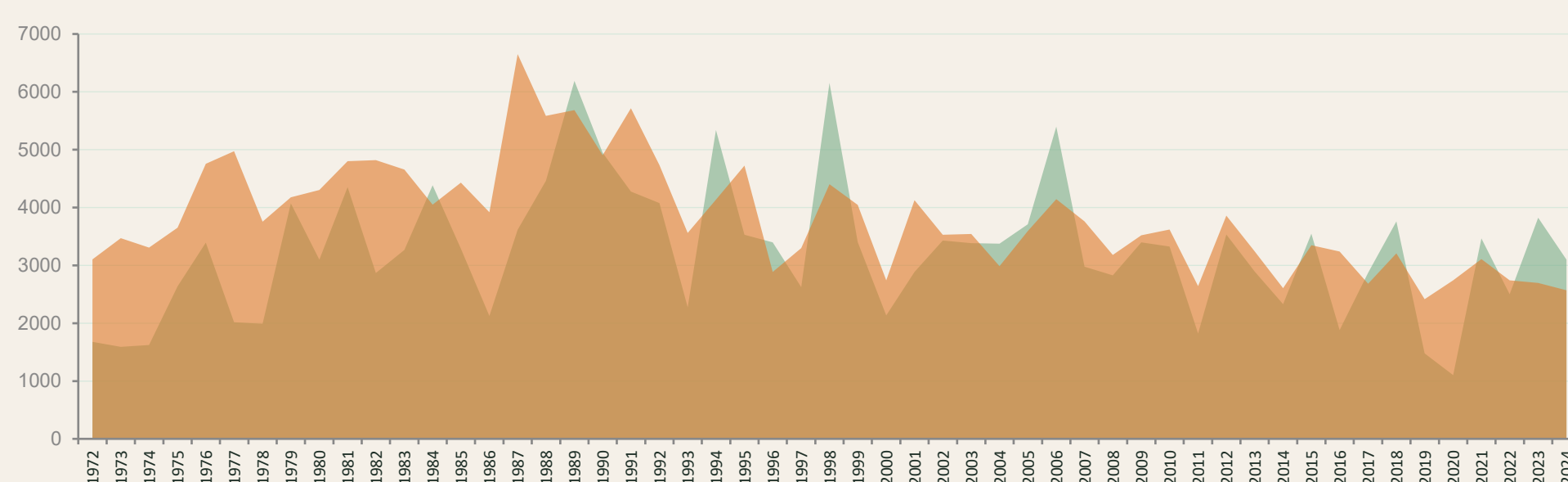


Fig. 6: Annual fire count by cause (1972–2024)

07 Class Distribution Analysis



By Month

Natural fires peak sharply in July–August (lightning season). Human fires have a distinct May spike and remain elevated through summer.

By Province

NS, NB, NL, and QC are heavily human-dominated. NT, YT, and PC are predominantly natural — remote boreal regions where lightning fires dominate.

By Decade

Human fires peaked in the 1980s and have declined, likely reflecting improved prevention. Natural fire counts are more variable, driven by drought and lightning cycles.

By Day of Week

Minimal weekend effect nationally — human fires do not spike on weekends, suggesting industrial and agricultural sources dominate over recreational causes.

08 Partial Dependence

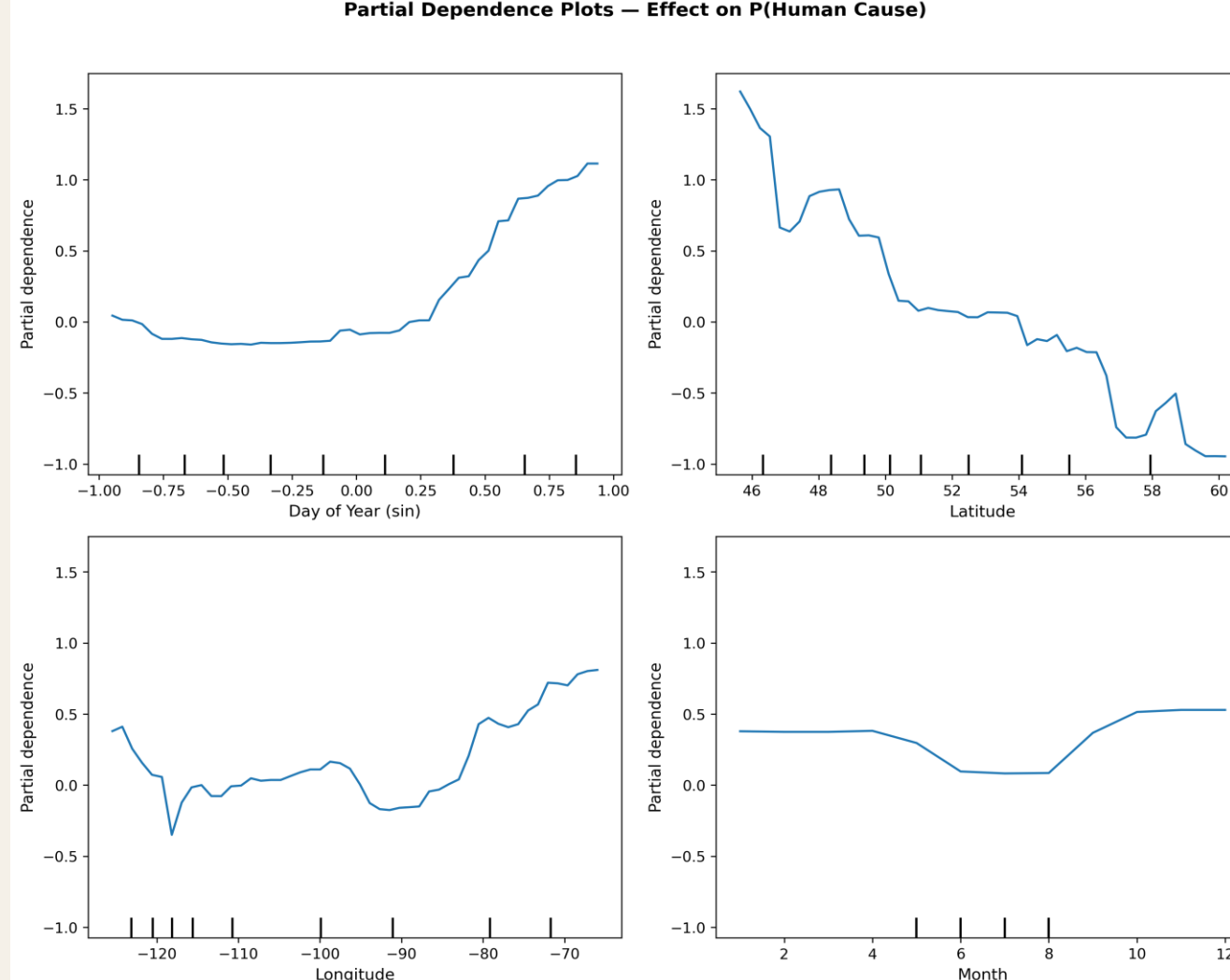


Fig. 8: Marginal effect of top 4 features on P(human cause)

Comparison to Prior Work

Rodrigues & de la Riva (2014) predicted human-caused wildfire *density* across 1×1 km grid cells in Spain — a spatial clustering task rather than per-fire cause classification. Using RF, BRT, SVM, and logistic regression on 93,573 grid cells (1988–2007), their best model (Random Forest) achieved an AUC of **0.746**. Our task is more granular: predicting the cause of each individual fire, nationally, with integrated climate context.

ROC-AUC vs. best prior model (Rodrigues RF, Spain 2014):

+18.6 pts

0.932 vs 0.746 · 374,037 fires vs 93,573

National scope · Integrated climate features · 4× larger dataset